

Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

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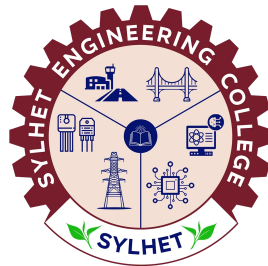
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Candidate's Declaration

This is to certify that the work presented in this thesis entitled, “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**”, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

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Recommendation Letter from Thesis Supervisor

The thesis entitled “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**” submitted by the students:

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is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor’s Degree.

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Certificate of Acceptance

The thesis titled “**Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)**” submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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The Authors

Abstract

Keywords:

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1 Introduction

The performance of flexible pavements is fundamentally governed by the quality of the bituminous binder and the physical characteristics of the constituent aggregates [6]. In the context of Bangladesh’s expanding road network, flexible pavements constitute the majority of the infrastructure, supporting rapid urbanization and increased traffic volumes. Bitumen acts as the primary binding agent, providing structural integrity and flexibility to withstand dynamic traffic loads and environmental stressors. However, the subtropical climate of Bangladesh, characterized by high temperatures and heavy monsoon rainfall, imposes severe demands on pavement materials, often leading to premature distress such as rutting and stripping [1].

To meet the growing demand for road construction, Bangladesh relies on a combination of locally produced and imported bitumen. The state-owned Eastern Refinery Limited (ERL) produces laboratory-grade bitumen, which is widely regarded for its consistency and adherence to standard specifications. Conversely, a significant portion of the market is supplied by imported low-cost bitumen to offset production shortfalls. The variation in the penetration grade and temperature susceptibility of these different binder sources plays a critical role in the long-term performance and climate resilience of the constructed pavements [2].

Despite the widespread use of imported bitumen, quality inconsistencies have raised significant concerns regarding its long-term viability. Empirical evidence suggests that adulterated or substandard imported bitumen contributes to rapid pavement deterioration and early rutting [5]. Consequently, agencies such as the Roads and Highways Department (RHD) enforce strict testing protocols and have occasionally restricted certain low-cost imported variants. In contrast, locally produced ERL bitumen has demonstrated superior performance in comparative evaluations and maintains a stable pricing structure, although systematic economic and mechanical trade-off analyses remain limited [3]. A critical gap remains in understanding the comparative life-cycle performance of these two primary binder sources under localized traffic and climatic conditions.

Furthermore, the mechanical response of bituminous mixtures is significantly dictated by the shape and gradation of the coarse aggregates. Prior investigations have demonstrated that replacing flaky and elongated aggregates with cubical aggregates enhances the interlocking mechanism, thereby increasing the Marshall stability and rutting resistance of the mix [7, 8]. While the influence of aggregate shape factors is well-documented [4], no prior study has systematically compared the synergistic effects of substituting conventional aggregates with specifically proportioned cubical aggregates across different bitumen sources in Bangladesh.

This study aims to systematically evaluate and compare the fundamental properties, mix design performance, and cost-effectiveness of local laboratory-grade bitumen and imported low-cost bitumen, focusing on the influence of coarse aggregate shape. The specific objectives include assessing standard binder properties, investigating the effect of substituting 20% of the coarse aggregate fraction (19 mm to 12.5 mm) with cubical aggregates on Marshall stability and flow, evaluating climate resistance via modified Marshall immersion, and performing a comparative economic analysis. The findings are expected to inform RHD specifications and promote sustainable road construction practices in Bangladesh. Section 2 reviews the existing literature; Section 3 details the experimental methodology; Section 4 presents the results and discussion; and Section 5 concludes the study with practical recommendations.

- 2 Literature Review**
- 3 Methodology**
- 4 Results and Discussion**
- 5 Conclusion**

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